

# Marketing Analytics

R Onboarding

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# What you will have at the end

- R and RStudio running — or a **Posit Cloud** account
- A course folder set up as an **RStudio Project** with subfolders for code, data, and graphics
- A script that **loads the course data** without errors
- A short **report knitted to HTML** — the format of your assignments

**Self-paced, about 2–3 hours, before session 2.** These slides summarize the onboarding module in the course book — the book has all details, screenshots, and the complete R code.

# Why R?

- **Free**, runs on every operating system
- A standard tool of data analysis in **research and industry**
- **Reproducible**: code can be re-run, checked, corrected, and shared — mouse clicks in a spreadsheet cannot

**AI writes R code cheaply** — that does not make R skills obsolete, it shifts them: the scarce skill is no longer *typing* code, but **reading and checking** it.

# Setting up

# Two routes to a working setup

## Route A (recommended): local installation

1. R from CRAN: <https://cran.r-project.org>
2. **RStudio Desktop**: <https://posit.co/download/rstudio-desktop/>

Install **in this order**, with the default settings. RStudio is built on top of R: R without RStudio works — RStudio without R does not.

## Route B (fallback): Posit Cloud

RStudio in the browser — no installation: <https://posit.cloud> → *New Project* → *New RStudio Project*.

Useful if the installation fails or your computer is managed by an employer. Free accounts have usage limits.

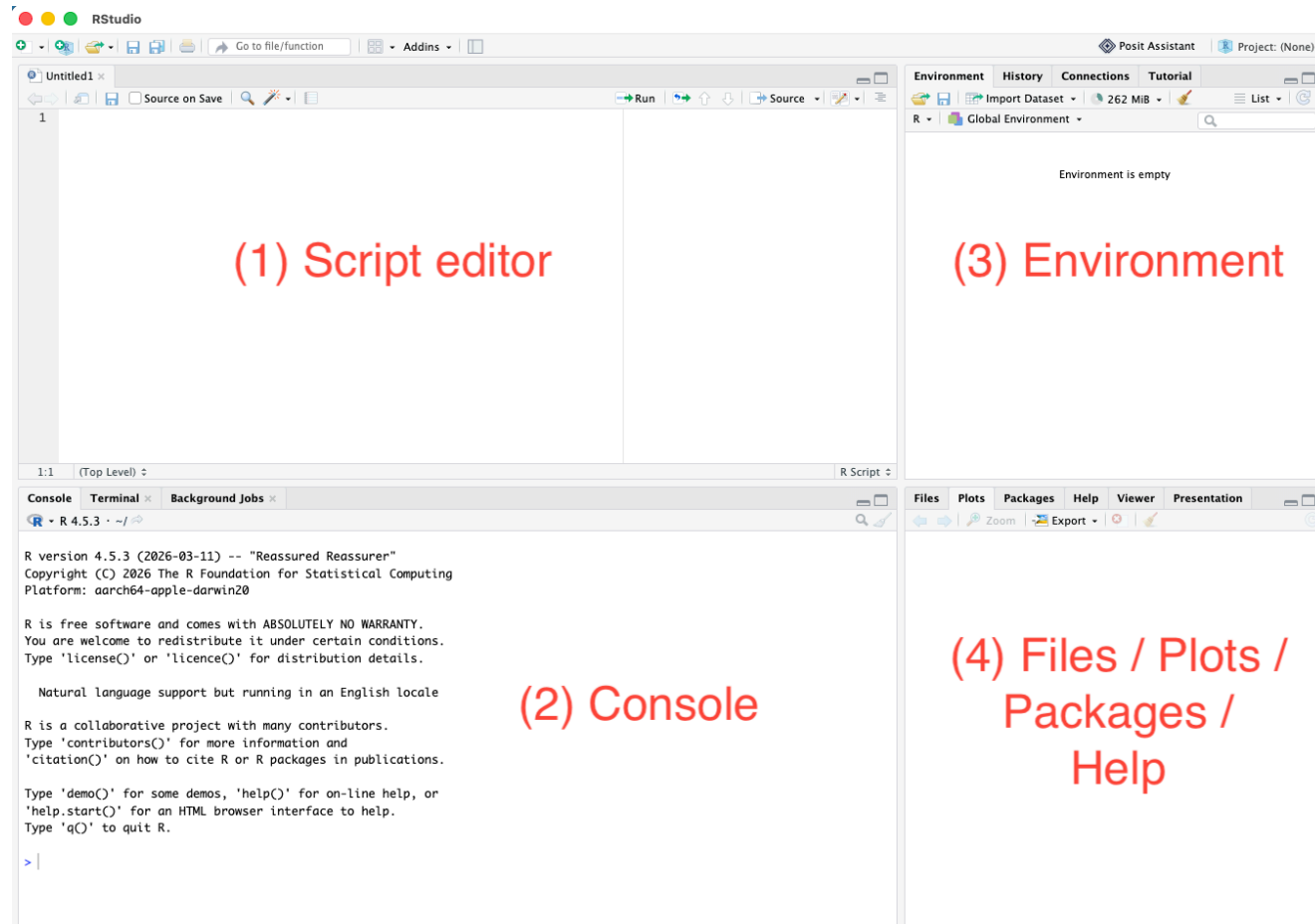
# Known pitfalls

- **Synced folders** (OneDrive, iCloud, Dropbox) cause strange errors → keep the course folder in a normal local folder
- **Special characters in paths** (umlauts, spaces) can break package installation on Windows → e.g., `C:/marketing-analytics`
- **macOS:** installation errors mentioning `xcrun` → run `xcode-select --install` in the Terminal app
- **Old R versions:** update — many packages need a recent R

**macOS:** “Operation not permitted” when knitting or reading files → RStudio lacks permission to access the folder.

*System Settings → Privacy & Security → Files & Folders (or Full Disk Access):* allow RStudio, then **restart RStudio** — or keep the course folder directly in your home folder (`~/marketing-analytics`).

# RStudio in ten minutes



## Pane

## Purpose

Script editor

*Write and keep your code (.R file)*

Console

*R executes code and prints results*

Environment

Objects in your current session

Files / Plots / Help

Files, charts, help pages

**Always write code in a script. Run a line with **Ctrl + Enter** (Mac: **Cmd + Enter**).**

# Work in an RStudio Project

*File → New Project → New Directory → New Project*

- R automatically looks for files in the **project folder** — no more “file not found”
- Start every work session by **opening the project** (`.Rproj` file)
- No `setwd("C:/Users/...")`: hard-coded paths break on every other computer

**Change one setting now:** *Tools → Global Options → General*

- “Save workspace to .RData on exit” → **Never**
- Untick “Restore .RData into workspace at startup”

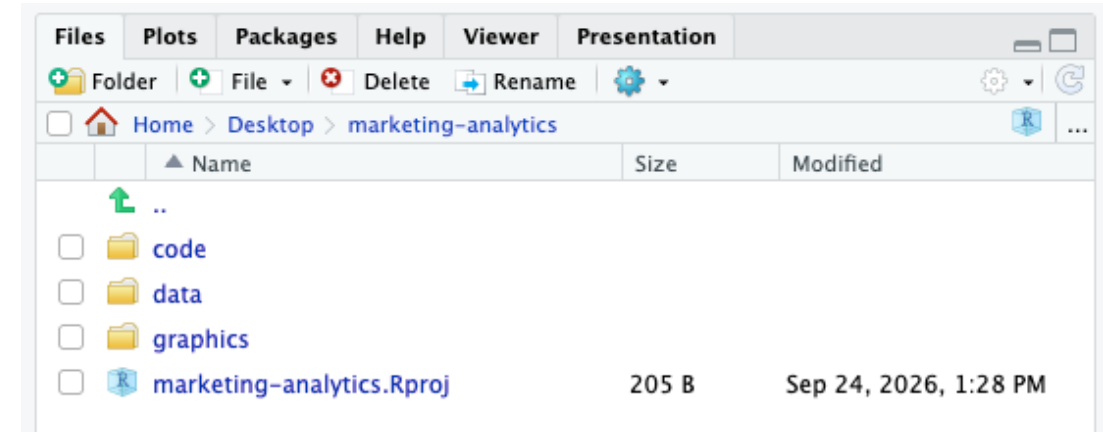
Your **script** is the record of your work — not the workspace.

# A tidy folder structure

```
marketing-analytics/      <- project folder
├── marketing-analytics.Rproj
├── code/                 <- .R and .Rmd files
├── data/                 <- .csv, .xlsx, .sav
└── graphics/             <- saved charts
```

Refer to files by their **relative path**:

- `read_csv2("data/music_data_fin.csv")`
- `ggsave("graphics/genre_shares.png")`



**Never edit raw data by hand · no spaces or brackets in file names: `assignment-1.Rmd`, not `Assignment 1 (final).Rmd`**

# First steps in R

# How to run code

1. Create a script: *File* → *New File* → *R Script* → save as `code/first-steps.R`
2. **Type** (or copy) the code into the script — typing it yourself helps you remember
3. Cursor in the line → **Ctrl + Enter** (Mac: **Cmd + Enter**) or the **Run** button; select several lines to run them together; whole script: **Ctrl/Cmd + Shift + Enter**
4. Compare the result in the **console** with the output in the book

**The console shows + instead of >?** R is waiting for you to finish a command — usually a missing closing bracket or quotation mark. Complete the command, or press **Esc** to abort.

# Objects and assignment

```
1 streams <- 163608      # assign a number to the object 'streams'
2 streams * 2            # objects can be used in calculations
```

```
[1] 327216
```

```
1 artist <- "Imagine Dragons" # text in quotation marks
2 artist
```

```
[1] "Imagine Dragons"
```

**<-** assigns (shortcut **Alt + -**, Mac: **Option + -**) · **#** starts a comment · R is case-sensitive: `streams` ≠ `Streams` · names: lower case with underscores, e.g., `mean_streams`

# Functions and arguments

```
1 seq(from = 1, to = 10, by = 3) # arguments named explicitly
```

```
[1] 1 4 7 10
```

```
1 seq(1, 10, 3) # same result, default order
```

```
[1] 1 4 7 10
```

```
1 round(3.14159, digits = 2)
```

```
[1] 3.14
```

Named arguments can come in any order · **help** for any function: `?seq` or **F1** on the function name · **Tab** shows completions while you type

# Vectors and data types

```
1 top_streams <- c(163608, 126687, 120480)
2 top_genres  <- c("Dance", "Alternative", "Latino")
3 mean(top_streams)
```

```
[1] 136925
```

```
1 class(top_genres)
```

```
[1] "character"
```

| Type      | Example                  |
|-----------|--------------------------|
| numeric   | 163608, 3.14             |
| character | "Dance"                  |
| factor    | factor(c("Pop", "Rock")) |
| logical   | TRUE, FALSE              |
| Date      | as.Date("2026-10-12")    |

Convert with `as.numeric()`, `as.character()`,  
`as.factor()`, `as.Date()`

# Data frames: R's tables

```
1 top_tracks <- data.frame(  
2   artist = c("Ed Sheeran", "Imagine Dragons", "J. Balvin"),  
3   streams = c(163608, 126687, 120480)  
4 )  
5 top_tracks$streams      # $ extracts one column
```

```
[1] 163608 126687 120480
```

```
1 str(top_tracks)      # structure: rows, columns, types
```

```
'data.frame':  3 obs. of  2 variables:  
 $ artist : chr  "Ed Sheeran" "Imagine Dragons" "J. Balvin"  
 $ streams: num  163608 126687 120480
```

# Packages

```
1 # once per computer:  
2 install.packages(c("tidyverse", "psych",  
3                   "readxl", "haven", "rmarkdown"))
```

```
1 # in every session, at the top of every script:  
2 library(tidyverse)
```

**install** = once per computer · **library** = every session

could not find function "..."? → the package is probably not loaded. Without loading:  
`dplyr::filter()`

# Three dialects of R

“Keep the pop tracks and compute their average number of streams”:

```
1 pop <- music_data[music_data$genre == "Pop", ]      # base R
2 mean(pop$streams)
3
4 music_data |> filter(genre == "Pop") |>             # tidyverse
5   summarise(mean_streams = mean(streams))
6
7 as.data.table(music_data)[genre == "Pop", mean(streams)] # data.table
```

| Dialect    | Strength                                    | Drawback              |
|------------|---------------------------------------------|-----------------------|
| Base R     | always available                            | hard to read          |
| tidyverse  | reads like a sentence; <code>ggplot2</code> | needs packages        |
| data.table | very fast                                   | cryptic for beginners |

**This course uses the tidyverse:** readable pipelines are easy to *check* — whether you or an AI wrote them. Be able to *read* base R.

# Keyboard shortcuts worth learning

| Action               | Windows              | Mac                 |
|----------------------|----------------------|---------------------|
| Run line / selection | Ctrl + Enter         | Cmd + Enter         |
| Run whole script     | Ctrl + Shift + Enter | Cmd + Shift + Enter |
| Assignment <—        | Alt + -              | Option + -          |
| Pipe  >              | Ctrl + Shift + M     | Cmd + Shift + M     |
| Insert code chunk    | Ctrl + Alt + I       | Cmd + Option + I    |
| Knit                 | Ctrl + Shift + K     | Cmd + Shift + K     |
| Comment lines        | Ctrl + Shift + C     | Cmd + Shift + C     |

**Native pipe:** *Tools → Global Options → Code → Editing*  
→ tick “**Use native pipe operator, |>**” — otherwise the shortcut inserts %>%.

**Own shortcuts:** *Tools → Modify Keyboard Shortcuts...*

# Data and reports

# Loading the course data

```
1 # directly from the course repository on GitHub (needs internet)
2 music_url <- "https://raw.githubusercontent.com/wu-rds/MA2026/main/data/music_data_fin.csv"
3 music_data <- read_csv2(music_url)
4
5 # or: download once into the project's data folder
6 music_data <- read_csv2("data/music_data_fin.csv")
```

```
1 dim(music_data) # number of rows and columns
```

```
[1] 66796    31
```

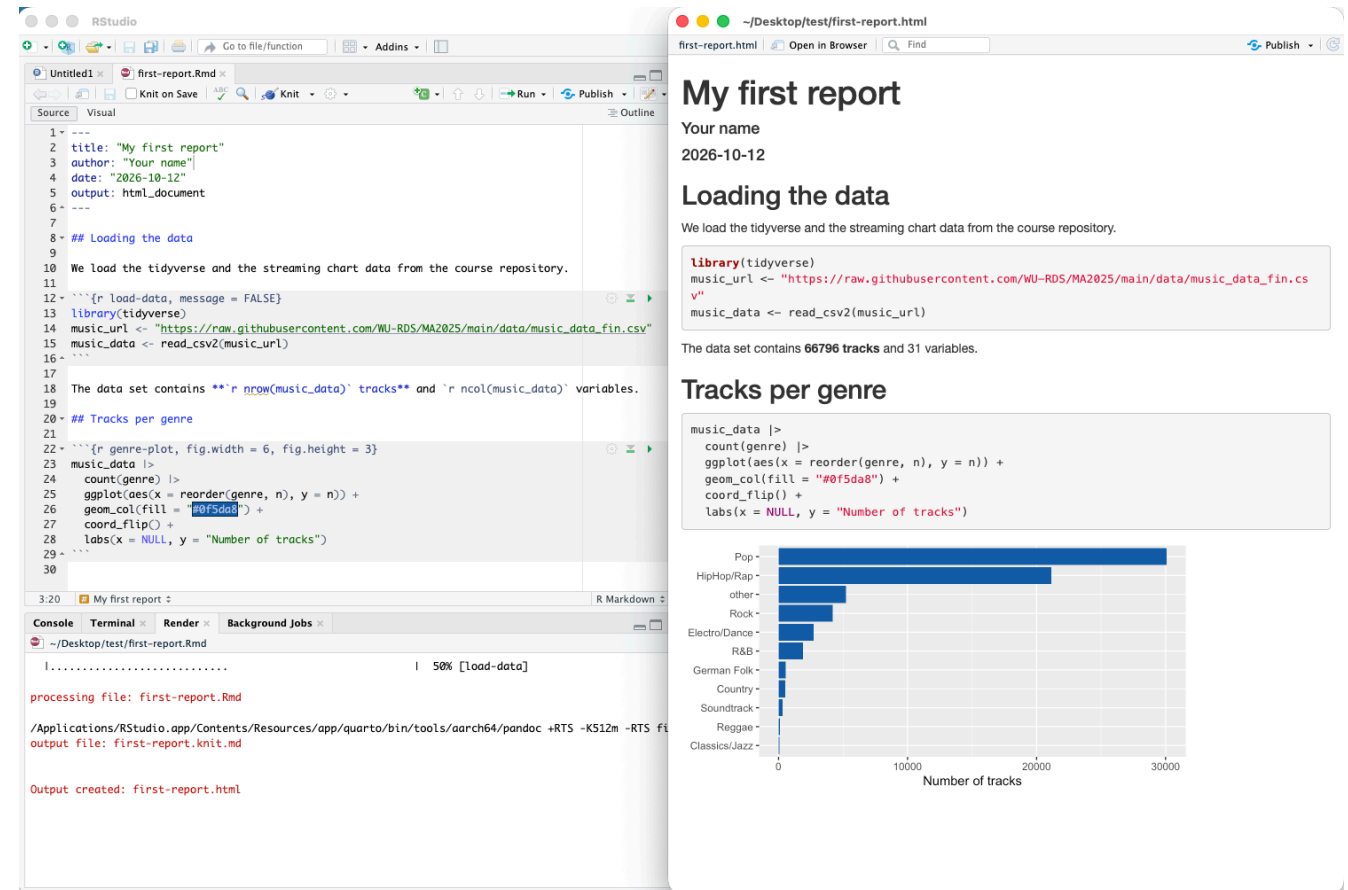
Why `read_csv2()` and not `read_csv()`? → first lesson of Chapter 2.

# Your first reproducible report

File → New File → R Markdown... → HTML → save as  
code/first-report.Rmd → Knit

Three ingredients:

- **Header** between `---`: title, author, output
- **Text** in Markdown: **`**bold**`**, *`*italic*`*, `# Heading`
- **Code chunks** between ````{r}` and `````
- **Inline code**: ``r nrow(music_data)`` puts results into the text



## Two rules that save hours

**The golden rule:** a report knits in a **fresh R session**. Everything it needs — `library()`, loading the data, every step — must be **in the report**. Works in the console but not when knitting? You rely on something that exists only in your session.

**Keep the chunk delimiters intact:** every chunk needs **three backticks** at the start (`` `{r}`) and at the end (`` ``). One deleted backtick → the report no longer knits. Insert chunks with the shortcut.

**File paths in reports:** a report in `code/` finds the data at `"../data/music_data_fin.csv"` (`..` = one folder up).

# Getting help — and working with AI

# Reading error messages

| Message                                         | What it usually means                                        |
|-------------------------------------------------|--------------------------------------------------------------|
| <code>could not find function "..."</code>      | Package not loaded — or function name misspelled             |
| <code>object '...' not found</code>             | Typo, wrong capitalization, or the creating line was not run |
| <code>cannot open file '...'</code>             | Wrong place or wrong name — are you in your RStudio Project? |
| <code>unexpected symbol/unexpected '...'</code> | Missing comma, bracket, or quotation mark                    |
| <code>there is no package called '...'</code>   | Not installed yet → <code>install.packages("...")</code>     |

Errors are not a sign that you are doing something wrong — they are R's way of telling you **where to look**. Read the message first.

# AI assistants: how they help

Chat assistants — ChatGPT, Claude, Gemini, Perplexity

Inside RStudio — GitHub Copilot suggests code as you type: *Tools* → *Global Options* → *Copilot* (GitHub account; free for verified students via GitHub Education)

```

149 ##
150 select(music_data, top10_artist_names, top10_track_streams, top10_track_explicit)
151
152
153 ## -----
154 select(music_data, -top10_track_release_date, -top10_track_explicit_1)
155
156
157 ## -----
158 # Create new variable as the log of the number of streams
159 music_data$log_streams <- log(music_data$top10_track_streams)
160 # Create an ascending count variable which might serve as an ID
161 music_data$obs_number <- 1:nrow(music_data)
162 head(music_data)
163
164
165 ## -----
166 music_data_new <- mutate(music_data,
167   sart_streams = sart(top10_track_streams),
168
169
159:1 (Untitled)
  
```

**Good at:** standard code (wrangling, charts, loops) · explaining functions and errors · debugging · suggesting alternatives

**Less reliable at:** statistical reasoning and interpreting results — fluent even when wrong

# Asking well

I am working in R with the following data frame:

```
> str(music_data)
'data.frame': 66796 obs. of 31 variables:
 $ isrc      : chr "BRRGE1603547" ...
 $ danceability: num 50.9 35.3 68.3 ...
 $ valence   : num 65.1 43.7 43.4 ...
```

Please provide the code for a publication-ready scatterplot between valence and danceability with pink hollow circles and a red regression line, and label the line with the regression coefficient.

- **Give context:** `str()` output, code, the *complete* error message
- **Be specific** about what you want
- **Start small**, then add complexity
- **Ask for explanations**, not just fixes

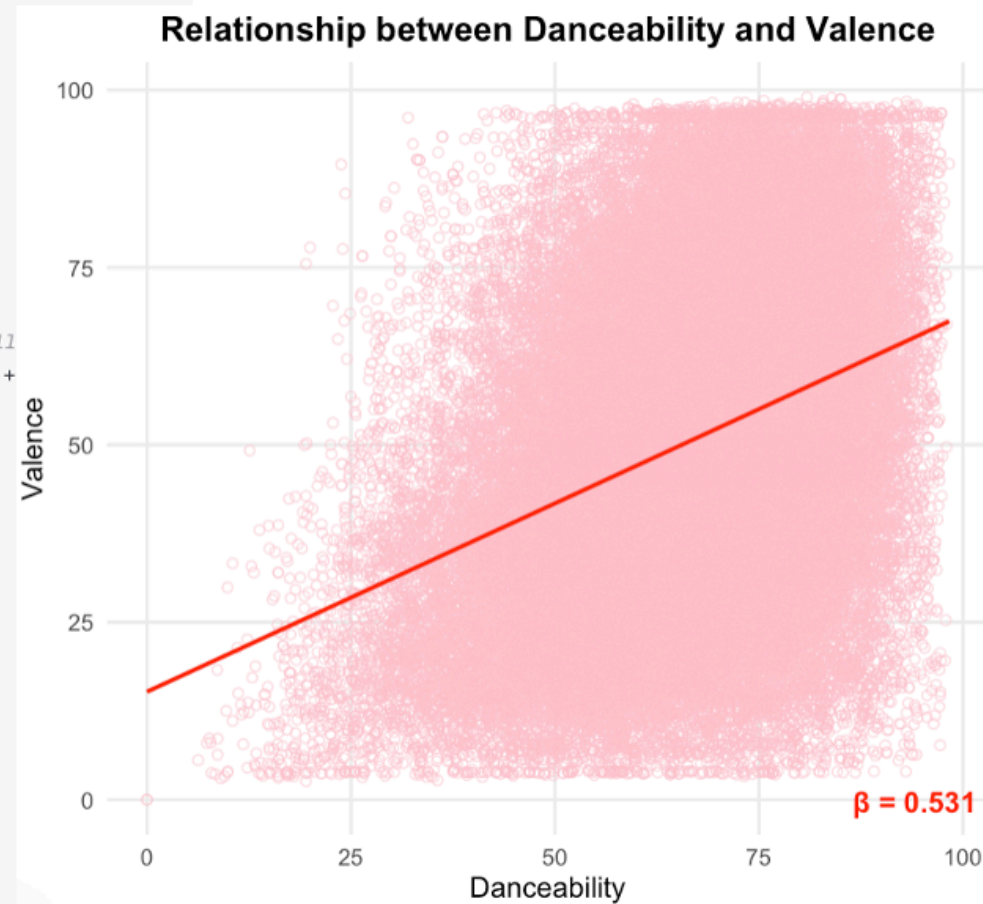
# Example: from prompt to chart

```
r
# Load necessary package
library(ggplot2)

# Fit linear model
model <- lm(valence ~ danceability, data = music_data)
coef_value <- coef(model)[2]

# Create scatterplot
ggplot(music_data, aes(x = danceability, y = valence)) +
  geom_point(color = "pink", shape = 1, size = 2, alpha = 0.6) + # holl
  geom_smooth(method = "lm", se = FALSE, color = "red", linewidth = 1) +
  annotate("text",
    x = Inf, y = -Inf,
    label = paste0("β = ", round(coef_value, 3)),
    hjust = 1.1, vjust = -1.1,
    size = 5, color = "red", fontface = "bold") +
  labs(
    x = "Danceability",
    y = "Valence",
    title = "Relationship between Danceability and Valence"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    panel.grid.minor = element_blank()
  )
)
```

Copy code



Debugging works too: could not find function "geom\_pont" + your code → the assistant spots the typo (`geom_point()`) and returns the corrected code.

# Always verify: assistants can be confidently wrong

An assistant explained R's percentile calculation correctly — and then reported the **median of 1, 2, 10** as **2.5**. Five seconds in R:

```
1 x <- c(1, 2, 10)
2 quantile(x, c(0.25, 0.5, 0.75))
```

```
25% 50% 75%
1.5 2.0 6.0
```

## When we request several percentiles at once

R uses the same formula for each  $p$ :

| Percentile | $p$  | $h$ | $j$ | $v$ | Quantile                       |
|------------|------|-----|-----|-----|--------------------------------|
| 25%        | 0.25 | 1.5 | 1   | 0.5 | $1 + 0.5 \cdot (2 - 1) = 1.5$  |
| 50%        | 0.5  | 2.0 | 2   | 0.0 | <b>2.0</b>                     |
| 75%        | 0.75 | 2.5 | 2   | 0.5 | $2 + 0.5 \cdot (10 - 2) = 6.0$ |

So actually the 50% quantile (median) is **2.0**, not 2.5 — my earlier example had a small typo in that line (thanks for catching it!).

## ✓ Summary

- For  $x = c(1, 2, 10)$  and  $type = 7$ :
  - 25% = 1.5
  - 50% = 2.0
  - 75% = 6.0
- The 50% (median) corresponds exactly to the middle observation (2), since  $h = 2$  and no interpolation is needed there.

If you'd like, I can show how the formula changes for other  $type$ s (e.g.,  $type = 2$  or  $type = 5$ ) to see how they differ in interpolation behavior. Would you like that?

+ Ask anything

What saved the day was not a better prompt, but **running the code and checking the result**.

# Using AI assistants mindfully

- **Foundations first** — you need to know what you are asking, and recognize a wrong answer
- **Critical thinking** — check outputs for correctness, efficiency, and reproducibility
- **Accelerate, don't bypass, learning** — if you could not explain the code to someone else, you are not done
- **Transparency** — disclose AI use in your assignments; be able to explain every line you submit

## Checkpoint: ready for session 2?

1. Open your RStudio Project — subfolders `code`, `data`, `graphics`
2. Download the course data into `data/`
3. New R Markdown file: one chunk that loads the tidyverse and the data
4. One sentence with the number of tracks — via **inline code**
5. A second chunk with `str(music_data)`
6. **Knit** to HTML

HTML opens and shows the structure of the data? **Your setup works.**

Stuck? Forum on Canvas, your AI assistant — and the **setup clinic** at the start of session 2.